SCI-NOI v0.1

SCI-NOI v0.1 r0.4 Shared Normative Core

Implementation-blind Stage B draft

Scientific owner: Grant Wilson

SCI-NOI_NORMATIVE_CORE v0.1/draft-r0.4

Not frozen. No implementation conformity, validation, calibration, physical-noise, covariance-completeness, significance, performance, readiness, or production claim.

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SCI-NOI v0.1 r0.4 Shared Normative Core

Document identity: SCI-NOI_NORMATIVE_CORE v0.1/draft-r0.4

Scientific owner: Grant Wilson

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Status: implementation-blind proposed final Stage B draft; scientifically freezeable conditional; not frozen.

The complete shared normative authority is the following six files in fixed order:

1. normative/NOTATION.md
2. normative/DEFINITIONS.md
3. normative/EQUATIONS.md
4. normative/ASSUMPTIONS.md
5. normative/REQUIREMENTS.md
6. normative/PREDICTIONS.md

They are byte-bound by SCI-NOI_NORMATIVE_MODULE_BINDING v0.1/r0.4 in NORMATIVE_MODULE_BINDING.json, binding-file SHA-256 5c6de3bd5180c9231c79cabe5f5918938571340a9f836f437962779e0410d55a.

The binding records the immutable SCI-NOI_AUTHOR_PACKET_MANIFEST v0.1/r0.18 digest b6f8e7252e7f61f4506899cb3e8e26cf939887bb48464852713f8ce81ac77ca0, the immutable SCI-NOI_OWNER_DECISIONS v0.1/r0.18 digest 272ac939b8a7109a123073b1a39fcdd7ac4129c603683ee81257b94ab2f55a0b, the accepted SCI-NOI_OWNER_DECISION_SUPPLEMENT v0.1/r0.3 digest 4b622de96e72860e544bf3699e3a131bc9cca5d2be0b0fe5bb41096327e977e5, and the final-completion SCI-NOI_OWNER_DIRECTIVE v0.1/r0.4 digest f155c1e70c1a8431bf4baf68f111960456ef24e9e77544837ff2efbecb86f423.

The r0.4 modules preserve stable r0.3 IDs and append only requirements 043-047 and predictions 018-022. They close enabled/disabled cardinality, atomic UNC membership and normalization, canonical exact balance arithmetic, the joint member-draw law and key identity, estimator-specific effective-information typing, and the fixed-scale versus full STD response boundary.

Any module byte change requires a new binding and regenerated documents. No rationale, ECS procedure, table, record, manifest, or PDF rendering supersedes the modules. The normative-core PDF appends their exact text in the stated order.

Authority and claim boundary

The r0.18 profiles and Registry objects remain immutable. Four complete r0.4 profile-successor records are proposed, not approved, not Registry-bound, and not evaluable. They create no action merely by having names. The superseding proposed freeze manifest binds the final scientific authority set but does not itself freeze it.

This draft establishes no implementation conformity, numerical validation, calibration, physical-noise validity, covariance completeness, Gaussian significance, achieved performance, readiness, freeze, production suitability, or production authorization.

Appendix A. Exact Bound Normative Modules

SCI-NOI v0.1 r0.4 Normative Notation

Module identity: SCI-NOI_NORMATIVE_NOTATION v0.1/r0.4 Scientific owner: Grant Wilson. Status: proposed final; not frozen.

- - Symbol | Exact meaning

- i,d,h,b,p | Exact retained occurrence, observation-scoped detector/channel, exact network, member, and frozen MAP row identities.
- Z_i^PTC,gamma_i,G_pi,a_pi | PTC signal, exact MAP-facing coefficient, frozen projection, and positive contribution G_pi gamma_i.
- C_p,Q_p | Exact contributing occurrences and unsigned normalization sum_i a_pi.
- beta_d,D_h^+ | Exact detector mass and {d: beta_d>0 and exact h(d)=h}.
- epsilon_bd,tau_h | Active detector sign and exact no-default tolerance in [0,1).
- (n_x,q_x) | Canonical reduced arbitrary-precision rational for design quantity x; q_x>0, gcd one, zero only (0,1).
- A | Exact full admitted assignment set for one observation/array plan.
- B_resolved,A_UNC | Exact resolved member-identity set and initial-UNC admitted set; equal in base v0.1.
- D_epsilon_b | Occurrence-diagonal expansion of the active assignment.
- M_b,m_MAP | NOI member and independently governed ordinary SCI-MAP signal.
- Vhat_cond,sigma_cond,zeta_cond | Conditional marginal second moment, its square-root scale, and unit-one standardized MAP.
- N_requested,N_resolved,N_completed,N_admitted | Requested, resolved, completed, and initial-UNC admitted counts.
- N_unique_sign,N_unique_orbit,r_sign,r_map | Distinct assignments, complement orbits, uncentered sign rank, and named projected rank.
- I_attempt | Execution-attempt identity, distinct from scientific identity.

SCI-NOI v0.1 r0.4 Normative Definitions

Module identity: SCI-NOI_NORMATIVE_DEFINITIONS v0.1/r0.4 Scientific owner: Grant Wilson. Status: proposed final; not frozen.

D-001 Operations and route

GEN, UNC, and STD are separate. The ordinary route inserts one active detector sign at the exact PTC-to-frozen-MAP boundary; NOI owns design and realization, MAP owns conforming frozen arithmetic.

D-002 Fixed method

Every consequential adjacent state is fixed; relearning or reselection creates another method and ensemble.

D-003 Population and scope

Base v0.1 has one ensemble per exact observation and TolTEC array. Complete, active, zero-mass, zero-total, singleton, and ambiguous states retain their approved typed dispositions.

D-004 Selected design

Independent symmetric candidates are first-accepted under every exact active-stratum condition, with positive cap, fail-closed exhaustion, replacement, valid duplicate draws, separate complement orbits, and no forced pair.

D-005 Disabled and enabled

Disabled/not-requested means N_requested=0 and no design or product. Enabled means a positive integer request and complete equal-count resolution; a smaller ensemble is never success.

D-006 Exact arithmetic

Design quantities use canonical reduced arbitrary-precision rational pairs and exact integer cross multiplication. This identity determines A and the probability law.

D-007 Joint law

Conditional on complete resolution, epsilon_b iid ~ Uniform(A). Candidate streams are disjoint across strata, members, and attempts by exact counter-domain separation.

D-008 Source-bearing target

The fixed parent may contain source and residual structure. The ensemble is not a source-free null, repeated physical noise, or calibrated tail law.

D-009 Initial UNC

The primary product is conditional_detector_sign_randomization_marginal_second_moment. Known zero target mean makes it equal to target-law marginal variance; it is not displayed as map/noise variance or total uncertainty.

D-010 Atomic UNC membership

For base UNC, A_UNC=B_resolved and every resolved member completes, passes policy, shares the common domain, and enters the estimator. No survivor normalization exists.

D-011 Effective information

Effective Monte Carlo information is estimator-, target-, domain-, dependence-, and calculation-specific. It is not inferred from counts, uniqueness, or ranks.

D-012 STD response

zeta_cond is nonlinear and data-dependent because numerator and scale descend from the same observed parent. A fixed-scale derivative is not the full procedure response.

D-013 Ownership and profiles

GEN owns completion; NOI owns use policy; VAL binds/evaluates. Immutable r0.18 profiles remain exact; changed r0.4 uses are unavailable until approved and bound successors exist.

D-014 Claim ceiling

No reciprocal ordinary product, significance, detection, conformity, validation, calibration, performance, readiness, freeze, or production claim is supplied.

SCI-NOI v0.1 r0.4 Normative Equations

Module identity: SCI-NOI_NORMATIVE_EQUATIONS v0.1/r0.4 Scientific owner: Grant Wilson. Status: proposed final; not frozen.

SCI-NOI-EQ-001 a_pi=G_pi gamma_i.

SCI-NOI-EQ-002 Q_p=sum_{i in C_p} a_pi.

SCI-NOI-EQ-003 beta_d=sum_p sum_{i in C_p,d(i)=d} a_pi; D_h^+={d: beta_d>0 and exact h(d)=h}.

SCI-NOI-EQ-004 abs(sum_{d in D_h^+} epsilon_bd beta_d) <= tau_h sum_{d in D_h^+} beta_d, 0<=tau_h<1.

SCI-NOI-EQ-005 M_b(p)=[sum_{i in C_p} a_pi epsilon_b,d(i) Z_i^PTC]/Q_p; the sign occurs exactly once only on Z_i^PTC, and every other MAP fact/gate is frozen.

SCI-NOI-EQ-006 m_MAP(p)=[sum_i a_pi Z_i^PTC]/Q_p only when independently governed SCI-MAP is scientifically realized; NOI cannot manufacture it by all-plus-one assignment.

SCI-NOI-EQ-007 R_b^fixed=A_MAP,Pi D_epsilon_b H_PTC^fixed; required response companions have identical membership/sign occurrence.

SCI-NOI-EQ-008 E_D[epsilon_bd]=0 for active d; E_D[M_b(p)|fixed parent/operator]=0 on exact available rows.

SCI-NOI-EQ-009 A_UNC=B_resolved; N_admitted=N_completed=N_resolved=N_requested>0.

SCI-NOI-EQ-010 Vhat_cond(p)=(1/N_resolved) sum_{b in B_resolved} M_b(p)^2 on the exact common all-member domain.

SCI-NOI-EQ-011 epsilon_b iid ~ Uniform(A) conditional on complete design resolution.

SCI-NOI-EQ-012 For rational left numerator L_h=n_L/q_L, total T_h=n_T/q_T, and tau_h=n_tau/q_tau, admission is exact integer comparison n_L q_T q_tau <= n_tau n_T q_L after nonnegative canonical reduction.

SCI-NOI-EQ-013 sigma_cond=sqrt(Vhat_cond) and zeta_cond=m_MAP/sigma_cond on the exact finite-positive compatible support.

SCI-NOI-EQ-014 delta zeta_fixed_scale=diag(1/sigma_cond) delta m_MAP, only with scale explicitly held fixed.

SCI-NOI-EQ-015 delta zeta=delta m_MAP/sigma_cond - m_MAP delta Vhat_cond/(2 sigma_cond^3); the second term requires separately authorized complete-procedure response.

SCI-NOI v0.1 r0.4 Normative Assumptions

Module identity: SCI-NOI_NORMATIVE_ASSUMPTIONS v0.1/r0.4 Scientific owner: Grant Wilson. Status: proposed final; not frozen.

SCI-NOI-ASM-001 Authority is the exact r0.18 packet, r0.3 owner supplement, r0.4 owner directive, and this byte-bound module set; process records add no science. SCI-NOI-ASM-002 Numerical GEN remains unavailable until exact PTC coefficient/QC, canonical rational design values, numerical coverage_cut, MAP admission, response, profiles, and every parent/plan fact exist. SCI-NOI-ASM-003 The real signal and every member share identical occurrences, operator, denominator, support, response, and gates. SCI-NOI-ASM-004 Independent symmetric candidate bits and exact content-bound disjoint counters supply the selected joint law; scheduling does not. SCI-NOI-ASM-005 First acceptance conditional on success is uniform on A; the finite cap changes resolution probability only. SCI-NOI-ASM-006 Candidate rejection is construction, not a member or failure. Admitted-member or policy failure fails base UNC. SCI-NOI-ASM-007 The fixed parent may contain source and residuals; no physical-noise or significance model is supplied. SCI-NOI-ASM-008 r0.18 profiles and Registry objects remain immutable. Proposed r0.4 successors are unevaluable pending owner approval and binding. SCI-NOI-ASM-009 Adjacent parents remain immutable; transforms and relearning require external authority and new generations. SCI-NOI-ASM-010 Full STD response is unavailable until the complete conditional-second-moment response is separately authorized.

SCI-NOI v0.1 r0.4 Shared Normative Module: Requirements

Module identity: SCI-NOI_NORMATIVE_REQUIREMENTS v0.1/r0.4

Scientific owner: Grant Wilson

Status: implementation-blind Stage B draft authority; content-bound by SCI-NOI_NORMATIVE_MODULE_BINDING v0.1/r0.4; not frozen.

shall is required; shall not is prohibited; not_applicable and unavailable are typed scientific states, not numerical sentinels.

SCI-NOI-REQ-001 GEN, UNC, and STD shall remain separate operations. None shall automatically realize, authorize, or mutate another.

SCI-NOI-REQ-002 The ordinary route shall be NOI-GEN/PTC-TO-FROZEN-MAP-CONDITIONAL-SIGN@1; signed outputs shall be NOI realizations, not ordinary MAP science products.

SCI-NOI-REQ-003 Every GEN method shall classify consequential adjacent state as fixed, rerun/relearned, not applicable, or unavailable. Different graphs shall not share an ensemble.

SCI-NOI-REQ-004 The coherence unit shall be one stable detector/channel in one exact observation. Its sign shall be constant across all admitted occurrences and ordered by canonical scientific identity.

SCI-NOI-REQ-005 Network balance shall use frozen positive coefficient mass beta_d. No network, array, or observation shall balance another, and beta_d shall not be promoted to another scientific role.

SCI-NOI-REQ-006 Base v0.1 shall produce one independent observation-level ensemble per exact observation and TolTEC array. It shall define no cross- observation randomized ensemble, randomized coadd, or multi-observation GEN route. Any future route requires the separate contract listed in the r0.4 owner supplement.

SCI-NOI-REQ-007 Every ordinary member shall satisfy SCI-NOI-EQ-005. The sign shall act only on Z_i^PTC, exactly once; all other MAP facts and gates shall remain frozen.

SCI-NOI-REQ-008 Inline and materialized application shall be equivalent only for identical equation, occurrence membership, assignment, frozen operator, and output domain. All-+1 arithmetic shall not manufacture an SCI-MAP product.

SCI-NOI-REQ-009 SCI-NOI-EQ-007 shall govern the fixed signal/response relation. Required response companions shall use identical membership and the modifier exactly once; missing response shall be unavailable without hidden signal subsetting.

SCI-NOI-REQ-010 The complete population, D_h^+, zero-mass detectors, zero-total strata, singleton active strata, and ambiguous positive-mass membership shall remain separately typed. No zero-denominator ratio shall be evaluated.

SCI-NOI-REQ-011 Zero-mass detectors shall be excluded from the active sign matrix but retained as typed facts. Zero-total strata shall be not_applicable. Ambiguous positive-mass membership and every singleton active stratum shall make the selected balance design unavailable; neither shall be silently exempted or cross-balanced.

SCI-NOI-REQ-012 The selected ODQ-102D design shall use exact tau_h in [0,1) without default, independent symmetric Rademacher candidates, first-accepted bounded rejection, a positive exact preregistered cap, fail- closed exhaustion, with-replacement draws, equal weight 1/N_resolved, scheduling-independent content-bound generator/keying, and no tolerance relaxation or best-failed fallback.

SCI-NOI-REQ-013 Every design shall bind an exact target probability law or deterministic finite measure; stable population/order; assignment alphabet; probabilities/weights; equality, duplicate, complement, and equivalence relations; requested/resolved/completed/admitted/unique counts; applicable ranks/null spaces; deterministic reconstruction identity; lifecycle, generation, completion, and failure. Every applicable field shall be preregistered before candidate or member output inspection. Unused fields shall be not_applicable.

SCI-NOI-REQ-014 RNG identity and version, opaque seed/key bytes, namespace, canonical serialization, stable ordering, observation/array/stratum/member/ attempt counters, domain separation, tolerance, retry cap, and rejected- candidate count shall be required only for a selected stochastic rejection family. For the selected family they shall be exact and plan-bound. Conditional on success, finite first-accepted rejection shall be uniform over the admitted vectors; the cap shall change resolution probability only.

SCI-NOI-REQ-015 Under every accepted design, exact duplicate equality and exact complement-orbit relation shall be represented separately. Under the owner-selected independent with-replacement rejection family, complement- related assignments shall remain distinct and no complement pair shall be forced.

SCI-NOI-REQ-016 Exact duplicates shall remain distinct draws. Products shall report all six N_* counts, r_sign, r_map, and use-specific effective information separately; none shall substitute for another.

SCI-NOI-REQ-017 The source-bearing conditional randomization ensemble shall carry D-005's objective and exclusions. Zero target-law mean shall not imply source removal from members, moments, or tails.

SCI-NOI-REQ-018 Every admitted assignment shall complete through the exact frozen operator. Any admitted-member failure shall fail the initial UNC ensemble; survivors and partial accumulators are diagnostic only.

SCI-NOI-REQ-019 An idempotent retry may retain scientific identity only for identical deterministic work with a new I_attempt. Changed scientific state shall be a replacement with new identity.

SCI-NOI-REQ-020 Each persistence plan shall select persisted, compact-regeneration, streaming-sufficient-statistic, or a separately identified composite and shall declare reconstruction, regeneration, sufficiency, unavailable later questions, lifecycle, completion, and failure.

SCI-NOI-REQ-021 An external deterministic transform shall require exact owner authority and parity for every operator fact; NOI shall not choose, tune, relocate, simplify, substitute, or silently omit it.

SCI-NOI-REQ-022 Wiener or FRUIT learning, selection, continuation, update, or per-member replay shall create every applicable new immutable owner, product, GEN, and UNC generation. Base numerical transformed routes remain unavailable.

SCI-NOI-REQ-023 Every UNC method shall bind exact target law, GEN membership, center, estimator, weights, dependence/missingness, domain, WCS/support/ response/unit, representation, ranks/null spaces, regularization, omissions, estimator uncertainty, lifecycle, claim, and named use.

SCI-NOI-REQ-024 Initial UNC shall consume only the exact resolved ensemble, with A_UNC = B_resolved and N_admitted = N_completed = N_resolved = N_requested > 0. Every resolved member shall complete, pass the member policy on the common domain, and enter the estimator. One ineligible, unavailable, incomplete, failed, or unrealized member shall make the complete initial UNC unavailable. Rejected candidates, survivors, subsets, mixed methods, failed persistence, and partial streaming state shall not be admitted or renormalized.

SCI-NOI-REQ-025 Initial UNC shall compute SCI-NOI-EQ-010 on the exact common domain with divisor N_resolved. It shall use the known zero target center and shall not subtract the finite ensemble mean, apply B-1, or replace the full resolved ensemble by a successful subset.

SCI-NOI-REQ-026 The primary product name shall be conditional_detector_sign_randomization_marginal_second_moment, with squared signal units. The equality to target-law marginal variance shall be stated separately; prohibited shortened names and physical-noise/total-uncertainty/ covariance/precision/significance claims shall not be used.

SCI-NOI-REQ-027 Numerical estimator uncertainty shall be explicit or unavailable. Counts, uniqueness, ranks, and effective information shall remain separate from that uncertainty.

SCI-NOI-REQ-028 Retained ensemble, marginal moment, projection, structured/full covariance, reciprocal, precision, weight, and unavailable representations shall remain separate roles. Missing covariance is unknown, not zero or independence.

SCI-NOI-REQ-029 No reciprocal shall enter the ordinary base bundle without a new exact owner disposition. Every reciprocal, inverse variance, precision, and consumer weight shall remain unavailable; STD shall consume sqrt(Vhat_cond) directly. Any future reciprocal shall use the exact name reciprocal_conditional_randomization_second_moment, finite-positive domain, separate identity/profile/use, and no prohibited promotion.

SCI-NOI-REQ-030 Initial STD shall use independently realized m_MAP, sigma_cond, and zeta_cond on the exact compatible finite-positive intersection. Missing MAP coefficient, numerical coverage_cut, MAP admission, or other required parent shall make m_MAP and STD unavailable. JINC remains separate and unavailable.

SCI-NOI-REQ-031 zeta_cond shall have unit 1 and the sole claim: "MAP signal standardized by the stated conditional detector-sign-randomization second-moment scale." No significance, probability, detection, false-alarm, completeness, purity, or catalog claim shall follow.

SCI-NOI-REQ-032 Admission shall retain separately named request, applicability, eligibility, and realization fields. Only the exact positive conjunction shall project to the exact named action.

SCI-NOI-REQ-033 Generation-input admission shall admit only one exact candidate occurrence for the selected GEN method; later operations remain separate.

SCI-NOI-REQ-034 UNC-member admission shall consume but not redefine GEN completion, failure, duplicate/equivalence, support, imprint, QC, persistence, lifecycle, cause, and provenance facts.

SCI-NOI-REQ-035 UNC-ensemble admission shall admit only one exact complete all-members-successful ensemble for one estimator/domain and shall not realize the estimator or authorize another use.

SCI-NOI-REQ-036 STD admission shall permit only the exact initial numerator/ scale join and claim. The proposed r0.4 successor profile has no authority by name alone; evaluation shall remain unavailable until exact owner approval and Registry/source binding.

SCI-NOI-REQ-037 Missing, conflicting, incomplete, unsupported, invalid, or failed required facts shall yield typed states. No undocumented sentinel, filename, shape, approximate WCS, or hidden default shall establish success.

SCI-NOI-REQ-038 PTC, MAP, JINC, transform, Wiener, FRUIT, and prior NOI parents shall remain immutable; later results shall be new versioned companions with exact ancestry and lifecycle.

SCI-NOI-REQ-039 No assignment, uncertainty, reciprocal, precision, scale, or weight shall become validity, support, exposure, or a PTC/MAP coefficient without separately approved successor authority.

SCI-NOI-REQ-040 Existing r0.18 profile and Registry objects shall remain immutable. The four complete proposed r0.4 successor records shall remain proposals, not executable authority. Affected evaluation shall remain unavailable until exact successor approval and Registry/source binding.

SCI-NOI-REQ-041 The rationale and ECS shall bind the exact six shared modules and shall not independently author equations, requirements, or predictions. Any module byte change requires a new binding and document generation.

SCI-NOI-REQ-042 This draft shall establish no implementation conformity, numerical validation, calibration, physical-noise validity, covariance completeness, Gaussian significance, achieved performance, readiness, freeze, production suitability, or production authorization.

SCI-NOI-REQ-043 N_requested = 0 shall mean only disabled/not requested: no design, members, successful empty ensemble, UNC, or STD exists. Enabled GEN shall require an exact positive integer N_requested; success shall require the complete identity N_resolved = N_requested > 0. Any member-assignment resolution failure within the preregistered cap shall fail the complete design, never realize a smaller successful ensemble. A rejected candidate is neither a member nor a member failure.

SCI-NOI-REQ-044 Every balance input a_pi, threshold tau_h, imbalance numerator and denominator, and admission comparison shall use canonical reduced arbitrary-precision rational pairs (n,q), with q > 0, gcd(|n|,q)=1, zero as (0,1), positive a_pi having n > 0, and 0 <= tau_h < 1. Serialization shall be canonical ASCII n/q; accumulation shall be exact in stable identity order; admission shall use exact cross multiplication. A missing exact rational shall make design resolution unavailable. Floating conversion shall not decide admission.

SCI-NOI-REQ-045 Conditional on complete design resolution, member assignments shall satisfy epsilon_b iid ~ Uniform(A). Candidate signs shall be independent symmetric Rademacher values before conditioning, with disjoint stratum streams before composition and disjoint member streams. Draws shall be with replacement; duplicates shall remain valid distinct draws; complement-related assignments shall be distinct and shall not be forced. The content-bound generator/keying record shall make outcomes independent of scheduling, process count, traversal, memory layout, and persistence choice.

SCI-NOI-REQ-046 N_resolved shall denote conditional Monte Carlo draws, not exposure, observations, or automatic effective sample size. Effective Monte Carlo information shall not be inferred from member count, unique count, duplicate count, complement-orbit count, or sign/map rank. Every effective- information product shall name its estimator, target, domain, dependence model, and calculation, or be unavailable.

SCI-NOI-REQ-047 Initial STD shall use sigma_cond = sqrt(Vhat_cond) and zeta_cond = m_MAP/sigma_cond. The fixed-scale derivative delta zeta_fixed_scale = diag(1/sigma_cond) delta m_MAP shall be labeled as a held-scale partial response, never the full response. The full response shall be delta zeta_cond = delta m_MAP/sigma_cond - m_MAP delta Vhat_cond / (2 sigma_cond^3) and shall remain unavailable until the second term has separate exact authority. Every response product shall carry dependence, response, support, and cause semantics; dividing a MAP response by sigma_cond shall not establish the full STD response.

SCI-NOI v0.1 r0.4 Shared Normative Module: Predictions

Module identity: SCI-NOI_NORMATIVE_PREDICTIONS v0.1/r0.4

Scientific owner: Grant Wilson

Status: implementation-blind Stage B draft authority; content-bound by SCI-NOI_NORMATIVE_MODULE_BINDING v0.1/r0.4; not frozen. These are analytic expectations, not implementation-conformity or empirical-validation reports.

SCI-NOI-PRED-001 A globally complemented active assignment has equal target probability under the approved complement-symmetric law.

SCI-NOI-PRED-002 Every active detector has target-law sign mean zero; the fixed linear member has target-law mean zero on each exact available row.

SCI-NOI-PRED-003 Complement symmetry does not remove source or structured residual contributions from individual members, second moments, or tails.

SCI-NOI-PRED-004 Complementing every active sign negates M_b while leaving the unsigned Q_p unchanged.

SCI-NOI-PRED-005 Complement-related members have identical pointwise squares and outer products while remaining distinct draws; exact duplicate and orbit relations remain separately represented.

SCI-NOI-PRED-006 Vhat_cond is nonnegative wherever available and is unavailable outside the exact common all-member domain.

SCI-NOI-PRED-007 Source-bearing fixed structure can appear in Vhat_cond; zero target-law mean does not prevent it.

SCI-NOI-PRED-008 Any failed admitted member makes initial UNC unavailable; completed survivors do not realize a partial estimator.

SCI-NOI-PRED-009 More members may improve estimation of the declared randomization target but cannot create exposure or reduce the immutable parent's realized noise as 1/sqrt(N).

SCI-NOI-PRED-010 A fixed owner-defined linear transform propagates a conditional moment under its exact operator; a learned per-member transform does not satisfy that identity by implication.

SCI-NOI-PRED-011 A positive profile decision neither realizes the next operation nor changes producer-owned facts.

SCI-NOI-PRED-012 Missing required response makes response-dependent output unavailable; signal membership does not silently shrink.

SCI-NOI-PRED-013 An idempotent retry may change attempt/evidence identity without changing scientific identity; a changed assignment or scientific state cannot.

SCI-NOI-PRED-014 Conditional on successful resolution, first-accepted finite rejection from independent symmetric candidates is uniform over admitted sign vectors: every admitted vector has the same base probability, and the common geometric probability of earlier rejection cancels in the conditional ratio. The cap changes success probability, not the accepted-vector law.

SCI-NOI-PRED-015 A zero-total-mass stratum is not_applicable to balancing and produces no imbalance ratio.

SCI-NOI-PRED-016 A singleton active stratum makes the selected balance design unavailable for every admissible tau_h; it is neither exempted nor cross-balanced.

SCI-NOI-PRED-017 zeta_cond has unit 1 on its available domain; dimensionlessness establishes no pivotal, significance, probability, detection, false-alarm, completeness, purity, or catalog interpretation.

SCI-NOI-PRED-018 A disabled request with N_requested = 0 produces no design or downstream product; an enabled successful request has exactly N_resolved = N_requested > 0; and failure to resolve any requested member within its cap produces a failed complete design, never a smaller successful ensemble.

SCI-NOI-PRED-019 If every resolved member completes but one member is ineligible for the exact common-domain UNC estimator, the complete initial UNC is unavailable; the remaining members are not renormalized into an estimator.

SCI-NOI-PRED-020 Exact reduced-rational cross multiplication admits equality at the balance boundary and rejects a value strictly above it, even when a finite floating representation rounds those two cases to the same comparison result.

SCI-NOI-PRED-021 Conditional on complete resolution, members are independent uniform draws from the admitted assignment set with replacement. Therefore an exact duplicate is a valid distinct draw, and a complement-related assignment is a distinct possible draw rather than a forced partner.

SCI-NOI-PRED-022 Holding Vhat_cond fixed removes the scale-response term and gives the stated fixed-scale partial derivative. When Vhat_cond responds to the perturbation, the full derivative contains the additional -m_MAP delta Vhat_cond/(2 sigma_cond^3) term; without separately authorized delta Vhat_cond, the full STD response is unavailable.